REPORT V1.1 ADDENDUM · TASKS 9–18

ASHA FISI JOB-TOOLKIT

Everything that changed since the 30-page baseline report: two newly unlocked sources, the corrected zero-transfer train whitelist, the project-review fix campaign, persistent job memory with daily digest, and the verified production numbers that supersede the baseline.

Addendum to: FISI Job-Search Toolkit — Phase 1–4 Report
Scope: Tasks 9–18 (Oct 2026) · status snapshot 2026-10-14
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SUPERSEDES · BASELINE CH· 3–5 · 2026-10-14

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1 Purpose and How to Read This Addendum

This addendum updates the delivered 30-page baseline report (Phase 1–4) with everything that changed during Tasks 9 through 18 of the build-out. The baseline remains accurate for source probing methodology, legal analysis, and the market overview; this document deliberately does not repeat any of it. Where the two disagree, this addendum wins — the single most important correction concerns the train-commute geography in baseline chapter 3, which turned out to be wrong in both directions and has been rebuilt from primary timetable data.

Chapters are ordered by impact rather than chronology. Section 2 covers the two newly unlocked sources, Section 3 the corrected transit whitelist, Section 4 the 19-finding code review and its fix campaign, Section 5 the persistent job memory and digest channel, Section 6 the open-source repository adoptions, and Section 7 the verified production numbers from the run that was executed while this addendum was being written. Section 8 is the updated operations runbook, which supersedes the quick-start instructions printed in the baseline appendix. All figures quoted here are reproducible from the repository: every run writes a manifest under `data/runs/` that links its outputs to per-source statistics, and the rejected-row audit continues to guarantee that no filtered posting disappears silently.

22	44	45/45	50
fetchable sources (was 20)	unique postings / run	unit tests passing	postings in job memory

2 Source Coverage: 20 to 22 Fetchable

Two more aggregators moved from the recipe column to production use since the baseline went out, and one previously mis-catalogued entry was corrected. The catalog now stands at 55 evaluated sources of which 22 are fetchable today: 18 live without credentials and 4 behind free API keys that Asha can register for herself (Adzuna, Jooble, Careerjet, WhatJobs). The fetch command is driven by the RUN_ORDER list in the source registry rather than by the catalog status column, so the catalog now states this explicitly to prevent future drift between documentation and behaviour.

Source	Baseline status	Status now	How it was unlocked
Stellenanzeigen.de	recipe (CSR shell, XHR undiscovered)	live (JSON API)	Browser session captured the results XHR during Task 11; replayed with curl_cffi Chrome-124 TLS impersonation, no proxy needed (39 raw / run)
Kimeta.de	recipe (Cloudflare, DC IP blocked)	live via proxy pool	SSR card recipe + validated free-proxy pool; silent degradation made visible by pool age signal (F12); 12/57 ceiling kept out of respect for robots.txt
Catalog counts	20 fetchable (16+4)	22 fetchable (18+4)	Task 17 review found README/email/handover quoting three different numbers; single source of truth is now the catalog CLI

Table 2-1 — Source coverage changes since the baseline report.

3 The Corrected Train-Commute Rule

The baseline geography chapter derived its 50-kilometre town list from road radius, not from the rail network. The Task 2-e analysis then verified every municipality against the official VGN GTFS timetable feed (j26-1, 2026-06-24) and found the hand-typed filter contradicted it in both directions: it accepted six towns that require a transfer since the December 2025 timetable change, and it missed roughly twenty towns that enjoy genuine zero-transfer service — including Coburg, which is about 65 km away but only 47 minutes by express train. In the last baseline-era production run, four of 42 rows (10%) violated the direct-train requirement.

The fix replaces the hand-typed expression with a data file: `data/geo/transit_whitelist.json`, transcribed from the verified analysis, from which all town-name regexes are generated at import time (umlaut-tolerant, separator-flexible). Every kept row now carries a **transit** column classifying its location: `0-xfer (NNmin)` for regular direct trains, `limited` for the two IC61-only towns (about two to four services per day), `borderline` for the 61–65-minute set, `1-xfer`, `none` for towns with no rail at all, and `>60min` for direct connections that fail the cutoff. Country-scoped sources are hard-filtered against the regular zero-transfer set; radius-scoped sources (BA, XING, kimeta, stellenanzeigen) keep every row but carry the label, so Asha can sort the two 1-transfer rows to the bottom instead of losing them.

Category	Examples	Old filter said	Verified verdict
Accepted wrongly	Neumarkt i.d.OPf., Feucht, Altdorf, Schwabach, Zirndorf, Winkelhaid	in radius (kept)	1 transfer — S2/S3 branches since the 2025-12-14 Rücktausch
Accepted wrongly	Herzogenaurach	in radius (kept)	no rail at all — bus-only town
Rejected wrongly	Coburg 47min, Lichtenfels, Haßfurt, Bamberg suburbs, Lauf/Ottensoos/Henfenfeld	outside radius (dropped)	regular 0-transfer, 16–60 min
Ambiguous pair	Höchstadt a.d.Aisch vs Hochstadt a.Main	both matched one regex	former has no rail; latter RE14 56 min — disambiguated by name suffix and PLZ

Table 3-1 — What changed under the corrected whitelist (full data: data/geo/transit_whitelist.json).

4 Project Review and the Fix Campaign

Task 17 audited all 2,412 lines of pipeline code and produced twenty findings: four correctness bugs, nine reliability gaps, and seven polish items. Task 18 then implemented every finding except one deliberately deferred scalability item. The three confirmed filter false-positives mattered most for Asha's daily results: a bare 450 match rejected a company mentioning 450 employees, the experience filter rejected a junior posting because the word *erfahrung* appeared inside the sentence “...erwartet wir nicht”, and any employer boilerplate mentioning Ausbildung could silently empty an entire source. All three are fixed with regression tests pinned to the exact sentences that triggered them.

ID	Fix shipped	Verified effect
F1	GTFS transit whitelist + transit column (Section 3)	4/42 violating rows now labelled; ~20 whitelisted towns no longer dropped
F2	Part-time 450-match requires currency marker	über 450 Mitarbeitern no longer rejects; 450-Euro/450€ still caught
F3	Experience rejection skips negated contexts (±90-char window)	Junior-friendly postings with "…erwartet wir nicht" kept, flagged negated-experience
F4	Ausbildung description-hits reject only role-shaped patterns	Boilerplate mentions get a note instead of silently dropping real jobs
F5	Post-merge filter hits land in the rejected audit	The nothing-disappears-silently promise holds for merged rows too
F6	sqlite-utils imported lazily	A wiped dependency can no longer take the whole CLI down (proven live)
F7/F9	requirements.txt + daily-script self-heal + flock	Fifth venv wipe restored from one file; overlapping runs locked out
F8	Catalog/README truth-out	One consistent count (22 fetchable); RUN_ORDER documented as authoritative
F10	Digest correlates expiry with run manifests	source-down no longer reported as posting-expired
F11	Persistent run manifests in data/runs/	Every output traceable to git head, per-source stats, pool state
F12	Proxy-pool age signal + fetch-log line	Kimeta degradation visible before it silently empties the run
F13	Memory: provenance merge, salary/PLZ/remote columns, url_key index	Re-titled reposts keep both sources; gap-fills instead of stale blanks
F14	Optional --parallel for non-WAF sources	Fetch-only in worker threads; wall-clock roughly halved on 17 sources
F15	tests/ package, 45 unit tests	45/45 green; every filter regression pinned with its triggering sentence
F16	snippet column (first 200 chars) in CSV	Excel triage without opening the JSON
F17	BA X-API-BLOCKED header watch + 429 retry	Early warning if the public BA key ever gets gated
F19	Stopword hygiene in the near-dupe scorer	Duplicate token removed; für/fuer/fu variants ignored

Table 4-1 — The Task 17 findings and their Task 18 fixes (F18 is this document; F20 deferred by the review).

5 Job Memory, Deduplication, and the Daily Digest

The pipeline is no longer a pile of CSVs. Every fetch upserts its results into a SQLite memory that keeps one row per unique posting with first-seen, last-seen, and times-seen, plus a full-text index over title, company, and city. The dedupe layer folds re-titled reposts across runs: when the same detail URL reappears under a different agency title, the memory row keeps both sources, the newest title, and any newly learned salary or location facts. Three dedupe layers run per fetch (exact key, canonical URL, near-title with company-token overlap), and suspiciously similar pairs are flagged for human review rather than auto-merged, after a Brunel-versus-FERCHAU case proved that 95% text similarity can still mean two distinct employers.

The daily digest turns the memory into a push message: what is new today, what most likely expired, and how large the memory has grown. Delivery runs through apprise, which speaks more than a hundred services — a Telegram bot or an ntfy topic needs only one environment variable (APPRISE_URL), and a healthchecks.io ping acts as a dead-man's switch that goes silent exactly when the daily run fails. The digest cross-checks its expiry claims against the run manifest of the day: postings whose sources all failed are reported as “not re-checked” rather than “expired”, which removes the false alarms that a cold proxy pool used to produce.

6 Open-Source Repository Integrations

Three rounds of repository scouting (87 repos surveyed in the baseline, 18 verified in the round-3 sweep, plus two later evaluations) followed one rule: adopt only what a live smoke test proved, defer with written triggers, and reject what duplicates existing capability. Five tools entered production service, one heavyweight was evaluated and deliberately deferred, and several popular options were rejected with evidence rather than opinion.

Repository	Verdict	What it does here
sqlite-utils (4.2)	adopted	the persistent job memory + FTS; 45 KB database carries all cross-run state
rapidfuzz (3.14)	adopted	near-dupe flagging; found a real missed duplicate on production data on day one
apprise (2.0)	adopted	digest push to 100+ notification services behind one URL
python-jobspy (1.3)	adopted	Indeed + LinkedIn guest-mode access inside the source registry
healthchecks.io hook	adopted	one-line dead-man's switch in the daily script
Scrapling (87k stars)	deferred	strongest parser-hardening candidate; trigger written: adopt when a WAF source breaks again
REA agents (208k stars)	deferred	reverse-engineering toolkit; triggers documented, no current backlog item fits
crawl4ai / browser tier	rejected	policy-green sources need no browser; agent-browser stays a manual unlock tool

Table 6-1 — Repository decisions with their evidence trail (research/legendary-repos/findings.md).

7 Verified Production Evidence (2026-10-14)

The numbers below come from the final verification run executed on this document's publication date, with every fix from Section 4 active. All 17 credential-free sources completed; the four key-gated sources await Asha's own free registrations, and LinkedIn's guest view returned an empty result set, which is its normal behaviour from a datacentre address. The run produced 44 unique postings after three-layer deduplication — 27 with confirmed full-time status, 17 flagged for a missing description, 4 carrying a junior signal. Eight hundred and one rejected rows are itemised in the audit file, including, for the first time, the post-merge rejections and the country-scope geography misses, so the quarterly whitelist review can be done from the audit alone.

The transit column classified 34 of the 44 rows as regular zero-transfer commutes, two as one-transfer (both via BA's server-side radius, kept and labelled by design), one as rail-less, and seven as unverified towns absent from

the GTFS analysis. Two previously lost postings reappeared precisely because the false-positive fixes landed: a junior first-level-support role at SIMBA DICKIE in Fürth and a system-administrator role at HEGMANNS in Nürnberg. The digest correctly identified four postings as likely expired — and, equally important, labelled nothing as expired that today's sources had merely failed to re-check. Job memory stands at 50 postings with 46 re-seen on the day.

Backup target	Retention	Contents
x0.at (V9xX.bundle)	~71 days	full v6 bundle, 174,370,824 bytes, sha256 verified via resume-loop download
litterbox.catbox.moe (rrubg4.bundle)	72 hours	same bundle, fast lane (~3.2 MB/s), Content-Length verified
filebin.net bin ashajobtoolkit-v6	6 days	human-friendly download; cookie-gated, two-step curl documented in BACKUPS.md
gofile	deregistered	datacentre blocked since v6; retry command kept in BACKUPS.md for home-IP use

Table 7-1 — Current off-site backups of the v6 bundle (refresh recommended after this addendum).

8 Updated Operations Runbook

The daily entry point now heals its own environment, which matters because the execution sandbox has wiped the virtual environment five times in as many days. One command runs the full chain: proxy-pool refresh (optional), environment check with automatic reinstall from requirements.txt, the fetch with hard filters and parallelism, the site mirror, the digest push, and the dead-man's-switch ping. The command refuses to start a second overlapping instance, and its log lives under logs/ next to a run manifest under data/runs/ that survives log rotation.

`bash scripts/daily_jobpull.sh [--with-proxies]` — daily run: self-heals the venv, fetches all 22 sources, mirrors outputs, pushes the digest

`/home/z/.venv/bin/python -m unittest discover -s tests -v` — 45 unit tests: filter regressions, whitelist pins, dedupe and memory behaviour

`python -m pipeline history new|gone|search|stats` — query the job memory (what is new today, what expired, full-text search)

`python -m pipeline sources` — the 55-source catalog with access recipes; RUN_ORDER is the fetch-authoritative list

`python -m pipeline fetch --sources ba,xing --fast` — targeted quick pull (skips per-job description enrichment)

Restore from scratch on any machine: clone the repository, create a virtual environment, run `pip install -r requirements.txt`, and execute the daily script — it recreates everything else it needs. The job memory travels inside the repository (data/jobs_history.db is committed deliberately), so restored instances keep their first-seen history and the digest remains accurate from day one. Quarterly maintenance is one line: re-run the GTFS script in

research/2-e and diff its output against data/geo/transit_whitelist.json before the December timetable change.